

Industry adoption of DevSecOps tools

The DevSecOps market size is expected to grow almost fourfold from US\$ 1.5 billion in 2018 to US\$ 5.9 billion in 2023, at a CAGR of 31.2 per cent. It is forecast to reach US\$ 6.5 billion by 2025, after growing at a CAGR of 28.85 per cent during 2020-2025 (see References at the end of the article).

According to a Synopsys report (<https://www.synopsys.com/glossary/what-is-devsecops.html>), organisations in multiple industries can implement DevSecOps to break down silos between development, security and operations so that they can release more secure software faster.

- **Automotive:** DevSecOps can reduce lengthy cycle times and help meet the software compliance standards.
- **Healthcare:** It can enable digital transformation efforts while maintaining the privacy and security of sensitive patient data as per regulations such as HIPAA.
- **Financial, retail, and e-commerce:** Here, DevSecOps can help to fix the Open Web Application Security Project, and also maintain data privacy and security compliance with PCI DSS payment card standards for transactions among consumers, retailers, financial services, etc.

DevSecOps solutions on cloud platforms are expected to help organisations deploy codes easily in the production process, along with enhanced IT security, high performance, and increased scalability.

Cloud and DevSecOps

Cloud and DevSecOps adoption by an organisation helps in providing agility, security, speed and quality to software processes. Companies can build applications in any programming language, as well as deploy and run them quickly and reliably on any infrastructure. Adoption of these technologies also supports automation of software release processes, faster application development, and better monitoring of applications and infrastructure performance.

Applications built on next-generation technologies include components such as omni-channel enablement, microservices adoption, API middleware, mobile apps, content management systems, etc. These applications need high availability and fault tolerance. The cloud can help to make the infrastructure or platform resources available within no time.

Cloud automation or Infrastructure as a Code should become a part of the culture of an organisation to eliminate manual activities in application installation and configuration.

DevSecOps reference framework

DevSecOps is a set or combination of tools that helps in the delivery, development and management of applications throughout a system's life cycle. At the organisation level, the software teams need to automate the entire cycle of build, provisioning and deployment of test environments, including the tools, scripts and test data to ensure rapid delivery. These teams need to collaborate around the application architecture and monitor event-based mechanisms for seamless data flow across the tool chains.

Listed below are the different stages any software or application has to pass through as part of the DevSecOps transformation journey:

- Portfolio management and collaboration
- Build
- Source code management
- Testing
- Continuous integration
- Deployment
- Configuration/provision
- Containerisation tools
- Repositories
- Database management
- Monitoring

The following sections briefly describe the different phases of the DevSecOps life cycle and the open source products that can be used for them.

Portfolio management: The current state of the application and future planning is considered at this stage. DevSecOps readiness assessment across the enterprise is done, along with requirements for DevSecOps implementation, and the approach for the development and transition into operations. The target stage is defined, and the transformation and execution plans are made. The ROI is calculated and the business strategy is made in this phase. In addition, identification of the initial DevSecOps process, the DevSecOps solution and its linkage to the cloud platform is done.

Build: DevSecOps establishes the interdependence of software development and IT operations, and helps an organisation produce software and IT services more rapidly, with frequent iterations.

Development of code may be done in any language, but is maintained by using version control tools. The most popular tools used are Git, SVN, SonarQube, Maven and Ant.

Source code management: Versions are maintained in a central repository that acts like a single source of truth. It helps developers to collaborate on the 'latest committed' code, and operations teams can access the same code when they plan to make a release. Whenever there is a fault during the release, Ops can quickly rollback the deployed code and revert to the previous stable state.